EC824A
Density Estimation

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Density Estimation – CDF

- Random sample $X_1, \ldots X_n$ from distribution F, with smooth density $f = F'$
- Interested in F and/or f
- X drawn from $F(X)$, each X_i is r.v. with distribution F,

$$F(x) = Pr(X \leqslant x)$$

  where $F(x)$ is the true distribution of each X_i evaluated at x
- If X continuous, $F(x)$ usually "S" shaped, from 0 to 1. Want to estimate F. Let

$$\widehat{F}(x) = \frac{1}{n}\sum_{i=1}^{n} \mathbb{1}(X_i \leqslant x) \quad (=: F_n(x))$$

  simply the fraction of data below x. $\widehat{F}(x)$: nonparametric estimator for $F(x)$

Density Estimation – CDF

- $\widehat{F}(x)$ is unbiased for $F(x)$ if averaged across infinite datasets equals $F(x)$ for every x
- $\widehat{F}(x)$ is unbiased for every x:

$$
\begin{aligned}
E[\widehat{F}(x)] &= E\left[\frac{1}{n}\sum_{i=1}^{n} 1(X_i \leqslant x)\right] = \frac{1}{n}\sum_{i=1}^{n} E\left[1(X_i \leqslant x)\right] \\
&= E\left[1(X_i \leqslant x)\right] = E\left[1(X \leqslant x)\right] \\
&= \int_{-\infty}^{\infty} 1(X \leqslant x) f(X) dX = \int_{-\infty}^{x} f(X) dX \\
&= F(X)|_{X=-\infty}^{x} = [F(x) - F(-\infty)] \\
&= [F(x) - 0] = F(x)
\end{aligned}
$$

- Note: not an averge over x – this is over r.v.'s X_i
- x is just an element of F, like an index: θ vís-a-vís θ_j

Density Estimation – CDF

- Calculate variance: (use $E[\widehat{F}(x)] = F(x)$). Let $\mathbb{1}_i := \mathbb{1}(X_i \leqslant x)$

$$Var(\widehat{F}(x)] = \mathbb{E}\left[(\widehat{F}(x) - F(x))^2\right] = \mathbb{E}\left[\left(\frac{1}{n}\sum_{i=1}^{n}\mathbb{1}_i - F(x)\right)^2\right]$$

- Assume X_i independent draws: $\mathbb{E}[\mathbb{1}_i\mathbb{1}_j] = \mathbb{E}[\mathbb{1}_i]\mathbb{E}[\mathbb{1}_j]$; note: $\mathbb{1}_i^2 = \mathbb{1}_i$

$$\mathbb{E}\Big[\Big(\frac{1}{n}\sum_{i=1}^{n}\mathbb{1}_i\Big)^2 + F(x)^2 - 2\frac{1}{n}\sum_{i=1}^{n}\mathbb{1}_i F(x)\Big] = \frac{1}{n^2}\sum_{i=1}^{n}\sum_{j=1}^{n}\mathbb{E}\mathbb{1}_i\mathbb{E}\mathbb{1}_i + F(x)^2 - 2\frac{1}{n}\sum_{i=1}^{n}\mathbb{E}[\mathbb{1}_i]F(x)$$

$$= \frac{1}{n^2}\sum_{i=1}^{n}\mathbb{E}\mathbb{1}_i^2 + \frac{1}{n^2}\sum_{i=1}^{n}\sum_{j\neq i}\mathbb{E}\mathbb{1}_i\mathbb{E}\mathbb{1}_i - F(x)^2 = \frac{1}{n}F(x)[1 - F(x)]$$

Density Estimation – CDF

- $\widehat{F}(x)$ is an average, so by CLT,

$$\sqrt{n}(\widehat{F}(x) - F(x)) \xrightarrow{d} N(0, F(x)[1 - F(x)])$$

- Hence $\widehat{F}(x)$ is "root-n-CAN": consistent asymptotic normal
- True at every point x

Density Estimation – Histograms

- Simplest density: univariate histogram, or bar-graph
- $X_1, \dots, X_n$ is an independent random sample from F with density f.
- Partition real line into $T = \{A_1, \dots, A_k\}$ disjoint sets

$$A(x) := \{A_i \in T | x \in A_i\}, \qquad \hat{f}_n(x) := \frac{\hat{F}_n\{A(x)\}}{\lambda\{A(x)\}} = \frac{\#\{X_j \in A(x)\}/n}{\text{length } (A(x))}$$

- E.g. $T_n : A_i = [(i-1)h + c, ih + c]$, and $\hat{f}_n(x) = (nh)^{-1}\#\{X_j \in A(x)\}$.
- Simply the count of observations in that bucket, normalised by the bucket width

Density Estimation – Histograms

- Define a population analogue of $\hat{f}_n(x)$ as

$$p_T(x) = \frac{P((A(x))}{\lambda(A(x))}$$

f continuous at x, let $h = \lambda(A(x))$ and $A(x) = [a, b]$

$$p_T(x) - f(x) = h^{-1} \int_a^b f(y)dy - f(x) = \frac{F(b) - F(a)}{h} - f(x) \to 0 \quad \text{as } h \to 0$$

where by mean value theorem, $(F(b) - F(a))/h = F'(\xi) = f(\xi)$ for $\xi \in [a, b]$.

Density Estimation – Histograms

- Popular measure of $\widehat{f}(x)$ performance is MSE ($Bias^2 + Var$):

$$MSE(\widehat{f}_n(x)) = \mathbb{E}[\widehat{f}_n(x) - f(x)]^2 = (\mathbb{E}[\widehat{f}_n(x)] - f(x))^2 + Var(\widehat{f}_n(x))$$

- Note $\mathbb{E}[\widehat{f}_n(x)] = \mathbb{E}[n^{-1}\sum_i I_A(X_i)/\lambda(A)] = n^{-1}\sum_i P(A)/\lambda(A) = p_T(x)$
- Variance term,

$$\begin{aligned} Var[n^{-1}\sum_i I_A(X_i)/\lambda(A)] &= (n\lambda(A))^{-2}\Big[\mathbb{E}\Big[(\sum_i I_A(X_i))^2\Big] - \mathbb{E}\Big[\sum_i I_A(X_i)\Big]^2\Big] \\ &= (nh)^{-2}\Big[\mathbb{E}\Big[(\sum_i I_A(X_i))^2\Big] - n^2P(A)^2\Big] \\ &= n^{-1}h^{-2}P(A)[1 - P(A)] \end{aligned}$$

- Hence $MSE = (\mathbb{E}[\widehat{f}_n(x)] - f(x))^2 + n^{-1}h^{-2}P(A)[1 - P(A)]$

Density Estimation – Histograms

- Saw bias goes to zero as $h \to 0$, hence $(\mathbb{E}[\widehat{f}_n(x)] - f(x))^2 \to 0$
- For variance: $n^{-1}h^{-2}P(A)[1 - P(A)] = (nh)^{-1}p_T(x)[1 - P(A)] \leqslant (nh)^{-1}p_T(x)$

Theorem 1

For $f(x)$ continuous with $h \to 0$ and $nh \to \infty$, MSE $\to 0$

- This result is pointwise (for each x) – would like something stronger, e.g.,

$$\int |\widehat{f}_n - f|dx = 2\int (f - \widehat{f}_n)^+ dx \to 0$$

- Equality from [Devroye (1987)] Section 2.5
- $(f - \widehat{f}_n)^+ < f$, f bounded by continuity, hence by dominated convergence theorem,

$$\int |\widehat{f}_n - f|dx \to 0$$

- "Standard" technique to strengthen pointwise convergence

Bandwidth Choice

- Want choice of bandwidth to minimise MSE

$$MSE(x) = (nh)^{-1} p_T(x) + o((nh)^{-1}) + (p_T(x) - f(x))^2$$

- $p_T(x) - f(x) = h^{-1} \int_a^b (f(y) - f(x)) dy$, and by mean value theorem, $\tilde{x} \in (a, b)$

$$h^{-1} \int_a^b (f(y) - f(x)) dy = h^{-1} \int_a^b (y - x) f'(\tilde{x}) dy$$

- Assume $|f'(\tilde{x})| \leqslant c_h(x)$ for $|x - y| < h$ (e.g. continuously differentiable),

$$|p_T(x) - f(x)| \leqslant h^{-1} \int_a^b |y - x||f'(\tilde{x})| dy \leqslant \int_a^b c_h(x) dy \leqslant h c_h(x)$$

where $\lim_{h \to 0} c_h(x) = |f'(x)|$

Bandwidth Choice

- To minimise with respect to h, use FOCs of MSE,

$$MSE(x) = (nh)^{-1}f + h^2(f')^2 + o((nh)^{-1}) + o(h^2), \qquad FOC: -(nh^2)^{-1}f + 2h(f')^2 = 0$$

- FOC implies $h = k \cdot n^{-1/3}$ for $k = (2(f')^2/f)^{-1/3}$
- Plug-in to MSE,

$$MSE(x) = n^{-2/3}[f/k + k^2(f')^2] + o(n^{-2/3})$$

  to find slower convergence rate than parametric problem (MSE of parametric models converged at n^{-1} rate)
- "Cost" of being nonparametric; of being flexible – will see A LOT about the bias-variance trade-off in second part of course

Bandwidth Choice

- What about points where $f'(x) = 0$?, e.g. peak of normal distribution
- For this consider an approximation to an integrated MSE

$$\int Var(\hat{f}_n)dx = \sum_{k=-\infty}^{\infty} \int_{A_k} Var(\hat{f}_n)dx = (nh)^{-1} \sum_{k=-\infty}^{\infty} P(A_k)[1 - P(A_k)]$$

where $\sum_{k=-\infty}^{\infty} P(A_k) = \int f dx = 1$. Mean value theorem,

$$\sum_{k=-\infty}^{\infty} P(A_k)^2 = \sum_{k=-\infty}^{\infty} f(\xi_k)^2 h^2 = h \int f(x)^2 dx + o(h)$$

where $P(A_k) = hp_T(x) = hf(\xi_k)$ for $\xi_k \in (a, b)$, $a \leqslant x \leqslant b$. Last equality by arbitrary closeness of integral to infinite discrete sum weighted by measure, $h = (b - a)$

Bandwidth Choice

- This gives the approximation,

$$\int Var(\hat{f}_n)dx = (nh)^{-1}\sum_{k=-\infty}^{\infty} P(A_k)[1-P(A_k)] = (nh)^{-1} - n^{-1}\int f(x)dx + o(n^{-1})$$

- Integrated bias, use $p_T(x)$ + mean value theorem ($\xi_k \in (a,b)$, $a \leqslant x \leqslant b$),

$$p_T(x) = f(\xi_k) = f(x) + (\xi_k - x)f'(x) + \frac{(\xi_k - x)^2}{2}f''(x) + \dots$$

- $(\xi_k - x)^2 = O(h^2)$ since $h = b - a$. Hence, if f'' bounded, for $c_k \in (a,b)$

$$\int_{A_k}(p_T(x) - f(x))^2dx = \int_{A_k}(\xi_k - x)^2(f'(x))^2 + O(h^4)dx$$
$$= O(h^2)f'(c_k)^2\int_{A_k} 1 + O(h^4)dx = (O(h^3) + o(h^3))f'(c_k)^2$$

Bandwidth Choice

- Hence, to form the integration $\sum_{k=-\infty}^{\infty}(O(h^3) + o(h^3))f'(c_k)^2$:

$$\sum_{k=-\infty}^{\infty}(O(h^3) + o(h^3))f'(c_k)^2 \leqslant O(h^3)\sum_{k=-\infty}^{\infty} f'(c_k)^2 = O(h^2)\int_{-\infty}^{\infty} f'(x)^2 dx + o(h^2)$$

- Asymptotic Integrated MSE (AIMSE):

$$AIMSE(h) = \frac{1}{nh} + O(h^2)\int_{-\infty}^{\infty} f'(x)^2 dx$$

- For bounded $\int_{-\infty}^{\infty} f'(x)^2 dx$, optimal $h^* = kn^{-1/3}$ again and $AIMSE(h^*) = O(n^{-2/3})$ (same as pointwise optimum for $f' \neq 0$)

Kernel Density Function

- Consider alternative estimate for $f(x)$, with $A = (x - h, x + h)$

$$\hat{f}_n(x) = (2hn)^{-1} \sum_i I_A(X_i) = \frac{F_n(x+h) - F_n(x-h)}{2h}$$

- $F^+ := F_n(x+h),\ F^- := F_n(x-h)$. Then $2nh\hat{f}_n(x) \sim Bin(n, p_n)$ with $p_n = F^+ - F^-$

$$\mathbb{E}\hat{f}_n(x) = \frac{p_n}{2h} \to f(x) \text{ as } h \to 0, \qquad Var[\hat{f}_n(x)] = \frac{p_n(1-p_n)}{4nh^2} \text{ as } (nh) \to \infty$$

  where Var goes to zero from $p_n/2h \to f$ and $(1 - p_n)/(nh) \to 0$ for $(nh) \to \infty$
- *Very* similar to histogram, except centered at x – has an important impact on bias!

Kernel Density Function

- If $X \sim Bin(n, p)$: $\mathbb{E}X = np$, $Var[X] = np(1-p)$, and DeMoivre Laplace:

$$Z_n = \frac{Y_n - np}{\sqrt{Var[Y]}} \xrightarrow{d} N(0,1)$$

- Hence, for $\Delta = F^+ - F^-$

$$Z_n = \frac{2nh\hat{f}_n(x) - n\Delta}{\sqrt{n\Delta(1-\Delta)}} = \frac{\sqrt{2nh}(\hat{f}_n(x) - (\Delta/2h))}{\sqrt{\Delta(1-\Delta)/2h}} \to \frac{\sqrt{2nh}(\hat{f}_n(x) - \mathbb{E}\hat{f}(x))}{\sqrt{f(x)}}$$

- Then, bias $(\mathbb{E}\hat{f} - f)$

$$\mathbb{E}\hat{f}(x) - f(x) = \frac{1}{2}\Big[\Big(\frac{F(x+h) - F(x)}{h} - f(x)\Big) + \Big(\frac{F(x) - F(x-h)}{h} - f(x)\Big)\Big]$$

Kernel Density Function

- Consider expansions of (continuous) CDFs around x, with $\eta_1, \eta_2 \in [0, 1]$

$$F(x+h) = F(x) + hf(x) + f'(x)h^2/2 + f''(x+\eta_1 h)h^3/6$$
$$F(x-h) = F(x) + hf(x) + f'(x)h^2/2 - f''(x-\eta_2 h)h^3/6$$

- Hence, for $h \to 0$, if $f''(x) < \infty$,

$$\mathbb{E}\hat{f}(x) - f(x) = \frac{h^3}{12}[f''(x+\eta_1 h) + f''(x-\eta_2 h)]/h \to \frac{h^2}{6}f''(x)$$

- First order $(hf'/2)$ cancelling out, big reduction in bias - going from order h to h^2
- Like histogram case, $Var[\hat{f}] \to f(x)/2nh$. Hence MSE, optimal MSE ($h^* = n^{-1/5}$),

$$MSE(x) = \frac{f(x)}{2nh} + \frac{h^4}{36}f''(x)^2 + o((nh)^{-1}) + o(h^4); \qquad MSE(x)^* = O(n^{-4/5})$$

- Still slower than parametric, but faster than $O(n^{-2/3})$

References

Devroye, L. (1987). *A course in density estimation*. Birkhauser Boston Inc.